

Finite Group Theory

Lectured by Kalyani Kansal

Scribed by Yu Coughlin

Autumn 2025

Contents

1	Finite groups	1
1.1	Conjugacy	1
1.2	Alternating groups	2
1.3	Matrix groups	2
1.4	Iwasawa's lemma	4
1.5	Sylow's theorems	4
1.6	Extensons and semidirect products	6
2	Series	6
2.1	Composition series	7
2.2	Solvable groups	8
2.3	Hall's theorem	9
2.4	Nilpotent groups	10
2.5	Solvable radical and fitting subgroup	12
3	Applications	12
3.1	The Frattini subgroup	12
3.2	Groups of order p^3	13
3.3	Groups of order 16	13
3.4	The transfer homomorphism	13

1 Finite groups

Proposition 1.0.1. If $A, B \trianglelefteq G$ are normal subgroups with $A \cap B = 1$ and $G = AB$, then $G \cong A \times B$.

Proposition 1.0.2 (Orbit-stabiliser). If a finite group G acts on a set Ω , then for all $\alpha \in \Omega$, the size of the G -orbit of α is the size of G divided by the size of the stabiliser.

Corollary 1.0.3. When G acts on itself by conjugation, for all $x \in G$, the size of the conjugacy class x^G is $|G : C_G(x)|$.

1.1 Conjugacy

Proposition 1.1.1. In S_n , conjugacy classes are sets of permutations with the same cyclic-shape.

Proof. If $c = (a_1 \ a_2 \ \cdots \ a_k) \in S_n$, then for all $g \in S_n$, we have $g c g^{-1} = (g(a_1) \ g(a_2) \ \cdots \ g(a_k))$. Now write $x \in S_n$ as a product of disjoint cycles, and we get $g x g^{-1}$ has the same cycle shap as x . $\square$

Corollary 1.1.2. If $x \in S_n$ has cycle chap $(r_1^{a_1}, \dots, r_k^{a_k})$, then $|C_{S_n}(c)| = \prod r_i^{a_i} a_i!$.

Proposition 1.1.3. Let $x \in A_n$, if $C_{S_n}(x) \leq A_n$, then $|x^{A_n}| = \frac{1}{2} |x^{S_n}|$, and x^{S_n} splits as two A_n -conjugacy classes; otherwise, $x^{A_n} = x^{S_n}$.

Proof. Either $C_{S_n}(x) = C_{A_n}(x)$, or $|C_{A_n}(x)| = \frac{1}{2} |C_{S_n}(x)|$, plug these values into orbit stabiliser. $\square$

Corollary 1.1.4. A_5 is simple.

Proof. The sizes of the conjugacy classes of A_5 are thus 1, 20, 12, 12, 15, the only union of these with order dividing 60 is 1 and A_5 . $\square$

By year 2 linear algebra we know every $A \in GL_n(k)$ is similar to a unique matrix in RCF, so conjugacy classes are given by these.

Proposition 1.1.5. If $H \leq G$, and Ω is the set of left cosets, then the map $G \rightarrow \text{Sym}(\Omega)$ by $g \mapsto [xH \mapsto gxH]$ is a transitive action with kernel

$$\ker(\pi) = N := \bigcap_{g \in G} gHg^{-1} \trianglelefteq G$$

Proof. The first part is immediate from the definitions, and we have $x \in \ker(\pi)$ iff $g^{-1}xg \in H$ for all $g \in G$ $\square$

Corollary 1.1.6. If $|G : N| = n$, then $|G/N| \leq \text{im } \pi \leq \text{Sym}(\Omega)$ has order $n!$

When we set $H = 1$, this recovers Cayley's theorem.

Definition 1.1.7. If $H \leq G$, then $N_G(H) = \{g \in G \mid gHg^{-1} = H\}$ is called the **normaliser** of H in G .

The number of distinct conjugates of $H \leq G$ is $|G : N_G(H)|$.

Proposition 1.1.8. If $H \leq G$, then $N_G(H) \cong C_G(H)$ is isomorphic to a subgroup of $\text{Aut}(H)$.

Proof. First isomorphism theorem on $\pi : N_G(H) \rightarrow \text{Aut}(H)$ by $n \mapsto [h \mapsto nhn^{-1}]$. $\square$

Proposition 1.1.9. If $H, K \leq G$ finite, then $|HK| = |H| |K| / |H \cap K|$ and if $h \leq N_G(K)$, then $HK \leq G$.

Proof. For the first part we count the fibres of the multiplication map $H \times K \rightarrow HK$. The second part is immediate from the definitions. $\square$

1.2 Alternating groups

Proposition 1.2.1. For $n \geq 3$, every element of A_n can be written as a product of 3-cycles.

Proof. Write $x \in A_n$ as a product of an even number of 2-cycles, $x = t_1 t_2 \cdots t_{2k}$, if t_1 and t_2 aren't disjoint, then their product is a 3-cycle, otherwise, up to relabelling we can write $t_1 t_2 = (12)(34) = (123)(234)$. $\square$

Theorem 1.2.2. For $n \geq 5$, the alternating group A_n is simple.

Proof. Do induction on n , we already know the base case $n = 5$. For $i = 1, \dots, n$ the stabiliser $(A_n)_i \cong A_{n-1}$, so if we have $N \trianglelefteq A_n$, then as $N \cap (A_n)_i \trianglelefteq A_{n-1}$ which, by induction, is simple, either $N \cap (A_n)_i = 1$ or $(A_n)_i$.

If $N \subset (A_n)_i$, then N contains a 3-cycle, and as the centraliser contains a disjoint transposition in $S_n \setminus A_n$, the A_n orbit or a 3-cycle equals the S_n orbit which is the whole cycle-shape, so N contains all 3-cycles, and thus is all of A_n .

If $N \cap (A_n)_i = 1$ for all i , then for all $1 \neq n \in N$, by relabelling we can write either $n = (12) \cdots$ or $n = (123 \cdots) \cdots$, both of which don't commute with $t := (124)$, so $[t, n] = (tnt^{-1})n^{-1} \in N$ is nontrivial, but this is $t(nt^{-1}n) = (123)(2jk)$ which moves at most 5 elements so is in some stabiliser, a contradiction, so $N = 1$. $\square$

1.3 Matrix groups

Proposition 1.3.1. If we write $GL_n(q) := GL_n(\mathbb{F}_q)$, for $q = p^m$ for some prime p and $m \geq 1$, then $|GL_n(q)| = (q^n - 1)(q^n - q) \cdots (q^n - q^{n-1})$.

Proof. A matrix in $GL_n(q)$ is invertible iff all of its rows are linearly-independent in $\mathbb{F}_q^n$. If we call the rows $r_1, \dots, r_n$: there are $|\mathbb{F}_q^n \setminus 0| = q^n - 1$ choices for r_1 , $|\mathbb{F}_q^n \setminus \text{Span}(r_1)| = q^n - q$ choices for r_2 , $|\mathbb{F}_q^n \setminus \text{Span}(r_1, r_2)| = q^n - q^2$ choices for r_3 , and so on, until there are $|\mathbb{F}_q^n \setminus \text{Span}(r_1, \dots, r_{n-1})| = q^n - q^{n-1}$ choices for r_n . $\square$

Proposition 1.3.2. $SL_n(q)$ is a normal subgroup of $GL_n(q)$ with order $|GL_n(q)| / (q - 1)$.

Proof. $SL_n(q)$ is the kernel of $\det : GL_n(q) \rightarrow \mathbb{F}_q^\times$. □

Definition 1.3.3. The **projective general linear group** $PGL_n(F)$ and **projective special linear group** $PSL_n(F)$ are

$$PGL_n(F) := GL_n(F)/Z(GL_n(F)) \quad PSL_n(F) := SL_n(F)/Z(SL_n(F))$$

It was a problem on the problem sheet to show these centres are

$$Z(GL_n(F)) = \{\lambda I_n \mid \lambda \in F^\times\} \quad Z(SL_n(F)) = \{\lambda I_n \mid \lambda \in \mu_n(F)\}$$

Proposition 1.3.4. If $A = GL_n(F)$ with $\det(A) = \mu$, then there exist U a product of elementary matrices, such that $A = U \operatorname{diag}(1, \dots, \mu)$.

Proof. This is basically just row-reduction from Year 1 linear algebra. If $A = (a_{ij})$, then by adding something to row 2 we have $a_{21} \neq 0$, and now $a_{21}^{-1}(1 - a_{11}r_2)$ to r_1 to get $a_{11} = 1$, do row reduction to the first row, and repeat by induction, this procedure never changes the determinant as elementary matrices have determinant 1. □

Corollary 1.3.5. For any field F and $n \geq 2$, $SL_n(F)$ is generated by elementary matrices.

Theorem 1.3.6. For $q > 3$, the group $PSL_2(q)$ is simple.

First proof. Write $G = SL_2(q)$ with $q > 3$, if $Z_{SL} \leq N \trianglelefteq G$, we show $N = G$. We do this in three parts.

Firstly, if $E_{12}(\lambda) \in N$ for some $\lambda \in \mathbb{F}_q^\times$, then $N = G$. Write $U_{12} = \{E_{12}(\beta) \mid \beta \in \mathbb{F}_q\} \cong (\mathbb{F}_q, +)$. For any $\alpha \in \mathbb{F}_q^\times$, we have

$$\begin{pmatrix} \alpha & 0 \\ 0 & \alpha^{-1} \end{pmatrix} \begin{pmatrix} 1 & \lambda \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \alpha^{-1} & 0 \\ 0 & \alpha \end{pmatrix} = E_{12}(\lambda\alpha^2) \in N$$

and hence $\{E_{12}(\lambda\alpha^2) \mid \alpha \in \mathbb{F}_q^\times\} \subseteq U_{12} \cap N$. The perfect squares in $\mathbb{F}_q^\times$ form a subgroup of size $(q-1)/\gcd(2, q-1)$. If we include the identity $E_{12}(0)$, then $|U_{12} \cap N| > \frac{1}{2}q$, and hence $U_{12} \leq N$. N will also contain

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 1 & \lambda \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = E_{21}(\lambda)$$

and hence N contains all elementary matrices, so $N = SL_2(q) = G$.

Secondly, if N contains some

$$A = \begin{pmatrix} \alpha & 0 \\ \beta & \alpha^{-1} \end{pmatrix}$$

for $\alpha \in \mathbb{F}_q^\times \setminus \{\pm 1\}$, then $N = G$. Indeed, as $N \trianglelefteq G$, N contains the commutator $[E_{21}(1), A] = E_{21}(1 - \alpha^{-2})$. So, by the previous step, $N = G$.

Finally, we can assume N contains no element of the form A . Let $n \in N \setminus Z_{SL}$. We assume n is not triangularisable, so its minimal polynomial is an irreducible $x^2 - \beta x + 1$ for some $\beta \in \mathbb{F}_q$. By RCF, there exists some $y \in GL_2(q)$ such that

$$y^{-1}ny = \begin{pmatrix} 0 & -1 \\ 1 & \beta \end{pmatrix}$$

and thus if we let $\mu = \det(y)$, and $y = g \operatorname{diag}(1, \mu)$ for $g \in SL_2(q)$, we get

$$g^{-1}ng = \begin{pmatrix} 1 & 0 \\ 0 & \mu \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & \beta \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & \mu^{-1} \end{pmatrix} = \begin{pmatrix} 0 & -\mu \\ \mu^{-1} & \beta \end{pmatrix} = n' \in N$$

and the commutator

$$\left[\begin{pmatrix} \alpha^{-1} & 0 \\ 0 & \alpha \end{pmatrix}, -n' \right] = \begin{pmatrix} \alpha^{-2} & 0 \\ \mu\beta(\alpha^2 - 1) & \alpha^2 \end{pmatrix} = n'' \in N$$

This matrix is of the form A , unless $\alpha^4 = 1$. We can always find such an α when $q \neq 5$.

If $q = 5$, then let $\alpha = 2$. Then $(n'')^2$ is a non-identity elementary matrix unless $\beta = 0$, in which case n is diagonalisable, a contradiction. □

We can also give a more general proof, and one that better generalises to showing lots of other matrix groups are simple.

1.4 Iwasawa's lemma

Definition 1.4.1. The action of a group G on a set Ω is called **2-transitive** if for all pairs $(x, y), (z, w) \in \Omega^2$, with $x \neq y$ and $z \neq w$, there exists some $g \in G$ with $g(x, y) = (z, w)$.

Lemma 1.4.2. The action of $SL_2(F)$ on $\Omega = \{\text{Span}(v) \mid v \in F^2 \setminus 0\}$ by $g \in SL_2(F)$ sending $\text{Span}(v)$ to $\text{Span}(g(v))$ is 2-transitive, and has kernel $Z = \{\pm I_2\}$.

Proof. Given $(\text{Span}(v_1), \text{Span}(v_2))$ and $(\text{Span}(w_1), \text{Span}(w_2))$ two pairs of distinct lines, v_1, v_2 and w_1, w_2 are both bases for F^2 . So there exists $g \in GL_2(F)$ sending $v_1 \mapsto w_1$ and $v_2 \mapsto w_2$. Take $h \in SL_2(F)$ by $v_1 \mapsto \det(g)^{-1}w_1$ and $v_2 \mapsto w_2$. $\square$

Corollary 1.4.3. The induced action of $G = PSL_2(F)$ on Ω is 2-transitive.

Theorem 1.4.4 (Iwasawa's lemma). Let $G \leq \text{Sym}(X)$ be a finite 2-transitive permutation group. If $G = G'$, and for all $\alpha \in X$ there exists some abelian $U \trianglelefteq G_\alpha$ such that $\langle gUg^{-1} \mid g \in G \rangle = G$, then G is simple.

Proof. Suppose we have $1 \neq N \trianglelefteq G$. Take $1 \neq n \in N$, and $\alpha, \beta \in X$ such that $n(\alpha) = \beta \neq \alpha$. By 2-transitivity, for all $\gamma \in X \setminus \alpha$ there exists some $g \in G$ with $g(\alpha, \beta) = (\alpha, \gamma)$. Then $gn g^{-1} \in N$ sends α to γ , thus N is transitive.

As N is transitive, for any $\alpha \in X$, we have $NG_\alpha = G$ and for all $g \in G$, there exists $n \in N$ with $n(\alpha) = g(\alpha)$, so $n^{-1}g \in G_\alpha$, and thus $g \in NG_\alpha$.

Let $U \trianglelefteq G_\alpha$ be as in the hypothesis, then as $N \trianglelefteq G$, we get $NU \trianglelefteq NG_\alpha = G$, this thus contains gUg^{-1} for all $g \in G$. By assumption, NU must be all of G .

We now just use the second isomorphism theorem to get $G/N = NU/N \cong U/(U \cap N)$ which is abelian, hence $G = G' \leq N$. Therefore $G = N$. $\square$

We can now give another proof that $PSL_2(q)$ is simple.

Second proof. Let e_1, e_2 be the standard basis of $\mathbb{F}_q^2$, and $\alpha = \text{Span}(e_1) \in \Omega$. All elements of G_α then must be of the form

$$G_\alpha = \left\{ \begin{pmatrix} a & b \\ 0 & a^{-1} \end{pmatrix} \mid a \in \mathbb{F}_q^\times, b \in \mathbb{F}_q \right\} / Z_{SL}$$

So if we take

$$U = \left\{ \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} Z \mid b \in \mathbb{F}_q \right\} \cong (\mathbb{F}_q, +)$$

then U is abelian and $U \trianglelefteq G_\alpha$. We can also take

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ b & 1 \end{pmatrix} \in U$$

so $|gUg^{-1}|$ contains all symmetric matrices and is thus all of $PSL_2(q)$.

Consider the commutator

$$\begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix} \begin{pmatrix} 1 & b \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a^{-1} & 0 \\ 0 & a \end{pmatrix} \begin{pmatrix} 1 & -b \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & b(a^2 - 1) \\ 0 & 1 \end{pmatrix}$$

as $q > 3$ we can always find $a \in \mathbb{F}_q \setminus \{0, \pm 1\}$, so G' contains all of U . As $G' \trianglelefteq G$, it also contains $\langle gUg^{-1} \rangle = G$. Hence, by Iwasawa's lemma, G is simple. $\square$

1.5 Sylow's theorems

Theorem 1.5.1 (Sylow 1). If $|G| = p^a m$ for p prime and $p \nmid m$, then G has a subgroup of order p^a .

Proof. We do induction on $|G|$. The base case $|G| = 1$ is trivial. First, for $x_1, \dots, x_k$ representatives of the distinct non-central conjugacy classes of G , we have the conjugacy class equation

$$|G| = |Z(G)| + \sum_{i=1}^k |x_i^G|$$

If $p \mid |Z(G)|$, then Cauchy's theorem gives some $z \in Z(G)$ of order p . As $\langle z \rangle \trianglelefteq G$, we can find $H \leq G/\langle z \rangle$ of order p^{a-1} . And $H = P/\langle z \rangle$ for some $P \leq G$, so $|P| = p^a$.

If $p \nmid |Z(G)|$, then one of the x_i^G has $p \nmid |x_i^G|$. As $|x_i^G| = |G : C_G(x_i)|$, we must have $p^a \mid |C_G(x_i)|$. Thus, by induction, there exists $P \leq C_G(x_i) < G$ with $|P| = p^a$. $\square$

Definition 1.5.2. Such a subgroup is called a **Sylow p -subgroup**. Write $\text{Syl}_p(G)$ for the set of all Sylow p -subgroups. G acts by conjugation on $\text{Syl}_p(G)$, the stabiliser of $P \in \text{Syl}_p(G)$ is $N_G(P)$. Write $n_p(G) := |\text{Syl}_p(G)|$.

Lemma 1.5.3. If $P \in \text{Syl}_p(G)$, Q is a p -subgroup of G , and $Q \leq N_G(P)$, then $Q \leq P$.

Proof. If $Q \leq N_G(P)$, then $PQ \leq G$, and $|PQ| = |P||Q|/|P \cap Q| = p^a |Q : P \cap Q|$, as Q is a p -group, and $p^{a+1} \nmid |G|$, we have $Q = P \cap Q$, thus $Q \leq P$. $\square$

Theorem 1.5.4 (Sylow 2). For all primes p , we have $n_p(G) \equiv 1 \pmod{p}$.

Proof. For $P \in \text{Syl}_p(G)$, consider the action of P on $\text{Syl}_p(G)$ by conjugation. If $\{Q\}$ is any singleton orbit, then $P \leq N_G(Q)$, so $P \leq Q$ and hence $P = Q$. So this action has exactly one singleton orbit. Now, by the orbit stabiliser theorem, the other orbits have size dividing $|P|$, so are divisible by p . As $n_p(G) = |\text{Syl}_p(G)|$ is a sum of orbits, $n_p(G) \equiv 1 \pmod{p}$. $\square$

Theorem 1.5.5 (Sylow 3). If Q is a p -subgroup of G , then there exists $P \in \text{Syl}_p(G)$ with $Q \leq P$.

Proof. Consider the action of Q on $\text{Syl}_p(G)$ by conjugation. As $n_p(G) \equiv 1 \pmod{p}$, not every Q -orbit has size divisible by p , so there is an orbit of size 1, call it $\{P\}$. Then $Q \leq N_G(P)$, so $Q \leq P$. $\square$

Theorem 1.5.6 (Sylow 4). The action of G on $\text{Syl}_p(G)$ by conjugation is transitive.

Proof. Let $P \in \text{Syl}_p(G)$, and Ω be the G -orbit of P in $\text{Syl}_p(G)$. Again, there is some orbit of size 1, thus $|\Omega| \equiv 1 \pmod{p}$. If $Q \in \text{Syl}_p(G)$, consider the action of Q on Ω . Again, there is some orbit of size 1, call it $\{R\}$. So $Q \leq N_G(R)$, hence $Q = R \in \Omega$. $\square$

Corollary 1.5.7. If $P \in \text{Syl}_p(G)$, then $n_p(G) = |G : N_G(P)|$. In particular, $n_p(G) \mid |G|$ and $n_p(G) = 1$ iff $P \trianglelefteq G$.

Proof. First, the stabiliser of P in $\text{Syl}_p(G)$ is $N_G(P)$, so this is just orbit stabiliser. The others follow. $\square$

Proposition 1.5.8. If $|G| = pq$ for $p > q$ primes, then G has a normal Sylow p -subgroup. If, furthermore, $q \nmid p-1$, then $G \cong C_{pq}$.

Proof. First, $n_p(G) \leq |G|/p = q$, and $n_p(G) \equiv 1 \pmod{p}$. As $q < p$, we thus have $n_p(G) = 1$. If $q \nmid p-1$, then $p \not\equiv 1 \pmod{q}$. So, similarly, we get a normal Sylow q -subgroup. Call these $P, Q \trianglelefteq G$. As $P \cap Q = 1$ and $G = PQ$, we must have $G \cong P \times Q \cong C_{pq}$. $\square$

Proposition 1.5.9. If $|G| = p^2q$ with p, q primes, then G has either a normal Sylow p -subgroup or a normal Sylow q -subgroup.

Proof. If $n_p(G), n_q(G) \neq 1$, then we can't have $n_q(G) = p$, as $p \not\equiv 1 \pmod{q}$, by $n_p(G) = q \equiv 1 \pmod{p}$. So $n_q(G) = p^2$. These must be p^2 disjoint copies of C_q , so $|G| - p^2$ elements of order q in G . This only leaves room for one Sylow p -subgroup. $\square$

Proposition 1.5.10. Let $|G| = p^a m$, where p is prime, $a \geq 1$, $m \geq 2$, and $p \nmid m$. If G is simple, then $|G| \nmid n_p(G)!$.

Proof. As G acts transitively on $\text{Syl}_p(G)$, there is a homomorphism $\phi : G \rightarrow \text{Sym}(\text{Syl}_p(G))$. The kernel is a proper normal subgroup of G , thus ϕ is injective. So G is isomorphic to a subgroup of $S_{n_p(G)}$. $\square$

1.6 Extensions and semidirect products

Definition 1.6.1. An extension of N by H is a group G with $N \cong N_0 \trianglelefteq G$ such that $G/N_0 \cong H$.

The Jordan-Holder theorem says that finding all extensions of N by H is the second half of classifying all finite groups. After we know the finite simple groups.

Definition 1.6.2. If G is an extensions of N by H , we say G is **split** if there is some $H_0 \leq G$ such that $G = N_0 H_0$ and $N_0 \cap H_0 = 1$. Note that $H = G/N_0 = N_0 H_0 / H_0 \cong H_0$.

Proposition 1.6.3. if $G = NH$ with $N \cap H = 1$ is a split extension, then we have the following:

1. every $g \in G$ can be written as $g = nh$ for unique $n \in N$ and $h \in H$;
2. for all $h \in H$ the map $\iota_h : N \rightarrow N$ by $\iota_h(n) = hnh^{-1}$ is in $\text{Aut}(N)$;
3. the map $\iota : H \rightarrow \text{Aut}(N)$ sending $h \mapsto \iota_h$ is a homomorphism;
4. multiplication is determined by ι as $(n_1 h_1)(n_2 h_2) = (n_1 \iota_{h_1}(n_2))(h_1 h_2)$.

Proof. These are all immediate. □

Definition 1.6.4. Given $\iota : H \rightarrow \text{Aut}(N)$, we give the set $N \times H$ the multiplication $(n_1, h_1)(n_2, h_2) = (n_1 \iota_{h_1}(n_2), h_1 h_2)$, where we write $\iota_{h_1} := \iota(h_1)$. Call this the **semidirect product** $N \rtimes_\iota H$.

Proposition 1.6.5. $N \rtimes_\iota H$ is a group under this multiplication, and a split extension of $N \times \{1\}$ by $\{1\} \times H$. If ι is the map from the previous proposition, then $G \cong N \rtimes_\iota H$.

Proof. These are also all immediate. □

Given to group N, H , if we find all homomorphisms $\iota : H \rightarrow \text{Aut}(N)$, and compute all the corresponding semidirect products $N \rtimes_\iota H$, then these are all the split extensions, up to isomorphism.

Proposition 1.6.6. Let N, H be finite groups, with $\iota, \kappa : H \rightarrow \text{Aut}(N)$ homomorphism. If there exists $\alpha \in \text{Aut}(N)$ and $\beta \in \text{Aut}(H)$ such that $\kappa(\beta(h)) = \alpha \iota(h) \alpha^{-1}$ for all $h \in H$, then $N \rtimes_\iota H \cong N \rtimes_\kappa H$.

Proof. Consider $\phi : N \rtimes_\iota H \rightarrow N \rtimes_\kappa H$ by $\phi(n, h) = (\alpha(n), \beta(h))$. This is an isomorphism. □

This condition is sufficient but certainly not necessary.

Proposition 1.6.7. If N and H are finite abelian groups of coprime order, then every extension of N by H splits.

Proof. Exercise. □

2 Series

Definition 2.0.1. A subgroup $N \leq G$ is **characteristic** if for all $\alpha \in \text{Aut}(G)$, $\alpha(N) = N$, written $N \text{ char } G$. These are clearly automatically normal.

Proposition 2.0.2. If $M \text{ char } N \trianglelefteq G$ are subgroups, then $M \trianglelefteq G$.

Proof. For all $g \in G$, conjugation by g is an automorphism of N , under which M is invariant. □

The centre, commutator, and inner automorphism group are all characteristic.

2.1 Composition series

Definition 2.1.1. A proper normal subgroup $N \triangleleft G$ is called **maximal** if there are no proper $N \subsetneq M \triangleleft G$.

Lemma 2.1.2. N is a maximal normal subgroup of G iff G/N is simple.

Proof. This is by the bijective correspondence between normal subgroups of G/N and normal subgroups of G containing N . $\square$

Definition 2.1.3. A **composition series** for a group G is a sequence of maximal subgroups

$$G = G_0 \triangleright G_1 \triangleright \cdots \triangleright G_{r-1} \triangleright G_r = 1$$

where the simple quotient G_i/G_{i-1} are called the **composition factors**.

Proposition 2.1.4 (Second isomorphism theorem). If $H \leq G$ and $N \trianglelefteq G$, then $H \cap N \trianglelefteq H$, $HN \leq G$, and $HN/N \cong H/(H \cap N)$.

Proof. The first two parts are clear, for the third consider the map $\phi : H \rightarrow HN/N$ by $\phi(h) = hN$, this is a surjective homomorphism and the kernel can be checked to be exactly $H \cap N$, the result now comes from the first isomorphism theorem. $\square$

Theorem 2.1.5 (Jordan-Hölder). The composition factors of a group are well-defined up to reordering.

Proof. Do induction on $|G|$, the base case is trivial. Suppose G has two composition series:

$$G \triangleright G_1 \triangleright G_2 \triangleright \cdots \triangleright G_r = 1 \quad G \triangleright H_1 \triangleright H_2 \triangleright \cdots \triangleright H_s = 1$$

then if $G_1 = H_1$, the two remaining series are composition series for a strictly smaller group, so have the same composition factors, by adding G/G_1 we are done.

Suppose, instead, that $G_1 \not\cong H_1$, then let $K = G_1 \cap H_1 \triangleleft G$ and choose a composition series

$$K \triangleright K_3 \triangleright \cdots \triangleright K_t = 1$$

I now claim $G_1/K \cong G/H_1$ and $H_1/K \cong G/G_1$, we have $G_1 \subset G_1H_1 \trianglelefteq G$, so $G_1H_1 = G$ and by the second isomorphism theorem $G/H_1 = G_1H_1/H_1 \cong G_1/(G_1 \cap H_1) = G_1/K$. As G_1/K is simple, we have the two composition series for G_1

$$G_1 \triangleright G_2 \triangleright G_3 \triangleright \cdots \triangleright G_r = 1 \quad G_1 \triangleright K \triangleright K_3 \triangleright \cdots \triangleright K_t = 1$$

which, by induction, must have the same length and composition factors, if we do the same for H_1 , then we can rewrite our initial composition factors as

$$\begin{aligned} \{G/G_1, G_1/G_2, G_2/G_3, \dots, G_{r-1}\} &= \{G/G_1, G_1/K, K/K_3, \dots, K_{t-1}\} \\ &= \{H_1/K, G/H_1, K/K_3, \dots, K_{t-1}\} = \{G/H_1, H_1/H_2, H_2/H_3, \dots, H_{s-1}\} \end{aligned} \quad \square$$

2.2 Solvable groups

Definition 2.2.1. A finite group G is **solvable** if it has a series of subgroups

$$1 = N_0 \leq N_1 \leq \dots \leq N_s = G$$

where $N_i \trianglelefteq G$ and N_i/N_{i-1} is abelian for all i .

Abelian groups are solvable. Finite p -groups are also solvable, as they always have nontrivial centre.

Definition 2.2.2. The n th **derived subgroup** $G^{(n)}$ of G is defined recursively as $G^{(0)} := G$ and $G^{(i+1)} := (G^{(i)})'$. The **derived series** of G is then the series $G \geq G^{(1)} \geq G^{(2)} \geq \dots$.

Proposition 2.2.3. If $\phi : G \rightarrow H$ is a surjective homomorphism, then $\phi(G^{(i)}) = H^{(i)}$.

Proof. For $x, y \in G$, we have $\phi([x, y]) = [\phi(x), \phi(y)]$, so ϕ being surjective is enough. $\square$

Corollary 2.2.4. For all i , we have $G^{(i)} \text{ char } G$.

Proposition 2.2.5. G is solvable iff $G^{(r)} = 1$ for some r .

Proof. If $G^{(r)} = 1$, then consider the derived series

$$1 = G^{(r)} \leq G^{(r-1)} \leq \dots \leq G^{(1)} \leq G$$

where each $G^{(i)} \trianglelefteq G$ and each quotient $G^{(i)}/G^{(i+1)} = G^{(i)}/(G^{(i)})'$ is abelian. So G is solvable.

Conversely, if G is solvable, with series

$$1 = N_0 \leq N_1 \leq \dots \leq N_s = G$$

where $N_i \trianglelefteq G$ and N_i/N_{i-1} is abelian for all i , then $N'_i \leq N_{i-1}$, so $G' \leq N_{s-1}$, $G^{(2)} \leq N'_{s-1} \leq N_{s-2}$ and so on. So, eventually, $G^{(s)} \leq N_0 = 1$. $\square$

Definition 2.2.6. For a solvable group G , the **derived length** of G is $\text{dl}(G) := \min\{r \mid G^{(r)} = 1\}$.

Lemma 2.2.7. Subgroups and quotient groups of solvable groups are solvable.

Proof. If G is solvable, and $H \leq G$, then $H' \leq G'$, and thus $H^{(i)} \leq G^{(i)}$. Hence $H^{(i)} = 1$ for some i , so H is solvable.

Now, if $N \trianglelefteq G$, and we consider the surjective quotient map $\phi : G \rightarrow G/N$, then $(G/N)^{(i)} = \phi(G^{(i)}) = 1$ for some i , hence G/N is solvable. $\square$

Lemma 2.2.8. If $N \triangleleft G$ and both N and G/N are solvable, then G is solvable.

Proof. $\square$

Proposition 2.2.9. If G is a finite group with composition series

$$G = G_0 \triangleright G_1 \triangleright \dots \triangleright G_{r-1} \triangleright G_r = 1$$

then G is solvable iff all the composition factors G_i/G_{i-1} are cyclic and of prime order.

Proof. $\square$

Definition 2.2.10. If A is a finite abelian group and there exists a prime p such that for all $x \in A$, $x^p = 1$, then we call A **elementary abelian**.

By the classification of finite abelian groups, the elementary abelian groups are $\mathbb{F}_p^n$.

Definition 2.2.11. A normal subgroup $N \trianglelefteq G$ is called **minimal normal** if N contains no smaller nontrivial normal subgroup of G .

Proposition 2.2.12. Let G is a finite group, with N a minimal normal subgroup. If N is solvable, then N is also elementary abelian.

Definition 2.2.13. For a finite group G , a **chief series** is a series

$$1 = N_0 \leq N_1 \leq \dots \leq N_r = G$$

where each $N_i \trianglelefteq G$, and each N_{i+1}/N_i is minimal normal in G/N_i . The quotient N_{i+1}/N_i are called **chief factors**.

While there is a Jordan-Hölder type theory for the uniqueness of the set of chief factors of a finite group, we won't prove it here.

Corollary 2.2.14. The chief factors of a finite solvable group are all elementary abelian.

2.3 Hall's theorem

Definition 2.3.1. Let π be a set of primes, we say a positive integer n is a π -**number** if all the prime factors of n are in π . In this context, write π' for the complement of π . A subgroup $H \leq G$ is called a π -**subgroup** if $|H|$ is a π -number.

Proposition 2.3.2 (Frattini Argument). Let $N \trianglelefteq G$ and $P \in \text{Syl}_p(N)$, then $G = N_G(P)N$.

Proof. For all $g \in G$, we have ${}^gP \leq {}^gN$, thus ${}^gP \in \text{Syl}_p(N)$ and by Sylow 4, there exists some $n \in N$ with ${}^gP = {}^nP$, thus $gn^{-1} \in N_G(P)$, and $g \in N_G(P)N$. $\square$

The following is a generalisation of a question on an earlier problem sheet. These are both special cases of the Schur-Zassenhaus theorem, which applies to arbitrary groups, and will not be proved here.

Theorem 2.3.3. If N and H be finite solvable groups of coprime orders, then every extension of N by H splits.

Theorem 2.3.4 (Hall 1). If π be any set of primes, and G a finite solvable group with $|G| = nm$ where n is a π -number and m a π' -number, then G has a subgroup of order n .

Definition 2.3.5. Such a subgroup H is called a **Hall π -subgroup** of G .

Like we have generalised Sylow 1 to Hall subgroups, there are similar generalisations of Sylow 3 and Sylow 4 to Hall subgroups, we won't prove these here.

2.4 Nilpotent groups

Definition 2.4.1. A finite group G is **nilpotent** if there exists a series of subgroups:

$$1 = G_0 \leq G_1 \leq \cdots \leq G_r = G$$

such that for all $0 \leq i \leq r-1$

$$G_i \trianglelefteq G \text{ and } G_{i+1}/G_i \leq Z(G/G_i)$$

such a series of subgroups is called a **central series**

Notably, all abelian groups are nilpotent; and all nilpotent groups are solvable, as $G_{i+1}/G_i \leq Z(G/G_i)$ implies G_{i+1}/G_i is abelian.

Proposition 2.4.2. Every finite p -group is nilpotent.

Proof. Let P be such a nontrivial p -group, define a series of subgroups as follows:

- $P_0 := 1$;
- $P_1 := Z(P)$;
- Have P_2 a subgroup containing P_1 such that $P_2/P_1 = Z(P/P_1)$;
- recursively define each P_{i+1} to be a subgroup containing P_i such that $P_{i+1}/P_i = Z(P/P_i)$.

as nontrivial p -groups always have nontrivial centre this series is strictly increase, so must terminate at some $P_r = P$. $\square$

Definition 2.4.3. The **upper central series** of a finite group G is the central series $Z_i(G)$ formed in the above proposition. We are taking the *biggest step* possible at each point.

Definition 2.4.4. For G a finite group and $H, K \subseteq G$, define

$$[H, K] := \langle [h, k] \mid h \in H, k \in K \rangle \leq G$$

one can immediately see $[H, K] = [K, H]$. We want to define the central series that takes the biggest step down at each point: this will be the **lower central series** of G defined as follows:

- $\gamma_0(G) := G$;
- For $\gamma_1(G)$ to be the next step down it must satisfy

$$G/\gamma_1(G) \leq Z(G/\gamma_1(G))$$

which just means $G/\gamma_1(G)$ is abelian, so we must have $[G, G] \leq \gamma_1(G)$, if we want to make the biggest just down $\gamma_1(G)$ must be as small as possible so we set $\gamma_1(G) := [G, G]$;

- Now given our $\gamma_i(G)$, to construct $\gamma_{i+1}(G)$ we must satisfy:

$$\gamma_i(G)/\gamma_{i+1}(G) \leq Z(G/\gamma_{i+1}(G))$$

so $\gamma_i(G)$ should commute with everything in G up to $\gamma_{i+1}(G)$ which is just:

$$[G, \gamma_i(G)] \leq \gamma_{i+1}(G)$$

so to continue with our biggest steps, we should set $\gamma_{i+1}(G) := [\gamma_i(G), G]$.

We should expect our two intuitively shortest possible sequences to be of the same minimal length, i.e. our group is nilpotent iff these series terminate. This is true, we will first need a lemma to prove it.

Lemma 2.4.5. For G a finite group and $X, Y \leq G$:

1. $Y \leq N_G(X)$ iff $[X, Y] \leq X$;
2. If $Y \trianglelefteq G$ and $Y \leq X$ then we have

$$X/Y \leq Z(G/Y) \iff [X, G] \leq Y$$

Proof. (1) $Y \leq N_G(X)$ iff for all $x \in X$ and $y \in Y$ we have $[x, y] = xyx^{-1}y^{-1} \in X$ iff $[X, Y] \leq X$.

(2) For all $x \in X$ we have: $xY \in Z(G/Y)$ iff $[xY, gY] = Y$ for all $g \in G$ iff $[x, g]Y = Y$ iff $[x, g] \in Y$. $\square$

This formalises our intuition of taking the biggest steps possible in the definitions of our upper and lower central series.

Theorem 2.4.6. For G a finite group and n a positive integer, $Z_n(G) = G$ iff $\gamma_n(G) = 1$. Furthermore, G is nilpotent iff these conditions hold for some n .

Proof. If either of the statements are true, then G is certainly nilpotent. So it suffices to prove for nilpotent G the set of n such that $Z_n(G) = G$ and $\gamma_n(G) = 1$ are equal and nonempty.

Let

$$1 = N_0 \leq N_1 \leq \dots \leq N_k = G$$

be a central series for G . By the previous lemma, for each i we have $[N_{i+1}, G] \leq N_i$.

Consider $\gamma_0(G) = G \leq N_k$, so $\gamma_1(G) = [G, G] \leq [N_k, G] \leq N_{k-1}$ and inductively:

$$\gamma_i(G) = [\gamma_{i-1}(G), G] \leq [N_{k-i+1}, G] \leq N_{k-i}$$

and $N_0 = 1$ we must have $\gamma_k(G) = 1$, applying this to our upper central series is half way.

Now consider $Z_0(G) = 1 = N_0(G)$, assume for $i \geq 1$ that $N_{i-1} \leq Z_{i-1}(G)$, equivalently:

$$[N_i, G] \leq N_{i-1} \leq Z_{i-1}(G)$$

moreover $[N_i Z_{i-1}(G), G] \leq Z_{i-1}$ as for all $n \in N_i$, $z \in Z_{i-1}(G)$, and $g \in G$ we have

$$[nz, g]Z_{i-1}(G) = [n, zg][z, g]Z_{i-1}(G) \subseteq [N_i, G]Z_{i-1}(G) \leq Z_{i-1}(G)$$

hence

$$N_i Z_{i-1}(G) / Z_{i-1}(G) \leq Z(G / Z_{i-1}(G)) = Z_i(G) / Z_{i-1}(G)$$

so $N_i \leq Z_i(G)$ and by induction we are done. $\square$

Definition 2.4.7. The a nilpotent group G the smallest integer n such that $Z_n(G) = G$ (or equivalently $\gamma_n(G) = 1$) is called the nilpotency class of G .

Corollary 2.4.8. Subgroups and quotients of nilpotent groups are also nilpotent.

Proof. Let G have nilpotency class n , if $H \leq G$ then $\gamma_n(H) \leq \gamma_n(G) = 1$ so H is nilpotent.

Let ϕ be the quotient map $G \rightarrow G/N$ for some $N \trianglelefteq G$, as homomorphisms commute with commutators and ϕ is surjective, we have $\phi(\gamma_i(G)) = \gamma_i(\phi(G))$ for all i , specifically $\gamma_n(\phi(G)) = 1$ so G/N is nilpotent. $\square$

If such an N and G/N are both nilpotent, G is not necessarily nilpotent. For example take $S_3 = C_3 \rtimes C_2$ which isn't nilpotent: $[S_3, S_3] = C_3$, $[C_3, S_3] = C_3$.

Nilpotent groups have lots of very nice characterisations.

Theorem 2.4.9. For a finite group G the following are equivalent:

1. G is nilpotent;
2. for any $H < G$, we have $H < N_G(H)$ is proper;
3. every maximal subgroup of G is normal;
4. every Sylow subgroup of G is normal;
5. $G \cong P_1 \times \dots \times P_k$ for P_i a p_i -group with all the $p_1, \dots, p_k$ distinct primes.

Proof. (1 $\Rightarrow$ 2) Let $1 = N_0 \leq \dots \leq N_k = G$ be a central series, and let $H < G$. Let r be the largest positive integer such that $N_r \leq H$, then $N_{r+1}/N_r \leq Z(G/N_r)$ and equivalently $[N_{r+1}, H] \leq N_r \leq H$, by our lemma we know this is equivalent to $N_{r+1} \leq N_G(H)$, hence $H \leq N_G(H)$.

(2 $\Rightarrow$ 3) Let M be a maximal subgroup of G , then $M \leq N_G(M) = G$, as M is maximal.

(3 $\Rightarrow$ 4) Let $P \in \text{Syl}_p(G)$ with P not normal in G , so $N_G(P) < G$. There exists a maximal subgroup M of G containing $N_G(P)$ with $M \triangleleft G$. As $P \in \text{Syl}_p(M)$ we can use the Frattini argument to get $G = N_G(P)M = M$, a contradiction, thus $P \trianglelefteq G$.

(4 $\Rightarrow$ 5) Let $P_1, \dots, P_k$ be the Sylow subgroups of G for distinct primes, as each P_i is normal in G , $[P_i, P_j] \leq P_i \cap P_j = 1$, hence $P_1 P_2 \dots P_k = G$ with each g written uniquely as a product of elements from different P_i .

(5 $\Rightarrow$ 1) Products commute with the γ_i , so the finite maximum of all the components nilpotency classes will be the nilpotency class of G . $\square$

2.5 Solvable radical and fitting subgroup

3 Applications

3.1 The Frattini subgroup

Definition 3.1.1. The G a finite group the intersection all all maximal subgroups is called the **Frattini subgroup** and denoted $\Phi(G)$.

Proposition 3.1.2. For all finite groups G and $x \in G$ the following are equivalent:

1. $x \in \Phi(G)$;
2. For all $S \subseteq G$, if $G = \langle S, x \rangle$, then $G = \langle S \rangle$.

Proof. (1 $\Rightarrow$ 2) Suppose $x \in \Phi(G)$ and there exists some $S \subset G$ such that $\langle S, x \rangle = G$ but $\langle S \rangle \neq G$, then there exists some maximal subgroup $M < G$ containing S , but $x \in \Phi(G) \leq M$, thus $G = \langle S, x \rangle \leq M$, a contradiction.

(2 $\Rightarrow$ 1) If $x \in G$ satisfies property 2 and $x \notin \Phi(G)$ then there exists some maximal $M < G$ with $x \notin M$, thus $\langle M, x \rangle = G$, but $\langle M \rangle \neq G$, a contradiction. $\square$

We call elements of $\Phi(G)$ the **non-generators** of G .

Proposition 3.1.3. $\Phi(G)$ is a characteristic nilpotent subgroup of G .

Proof. An automorphism of G must permute the maximal subgroups, so fixes their intersection, thus $\Phi(G)$ char G .

Let p be a prime and $P \in \text{Syl}_p(\Phi(G))$, by the Frattini argument, we must have $G = N_G(P)\Phi(G)$. So by the previous proposition (and some induction) $G = N_G(P)$ so $P \trianglelefteq G$, and a fortiori $G \trianglelefteq \Phi(G)$. Now, by the characterisation of nilpotent subgroups, $\Phi(G)$ is nilpotent. $\square$

Theorem 3.1.4 (Burnside Basis Theorem). For P a p -group with $|P| = p^n$ and $|\Phi(P)| = p^{n-d}$ for some $d \geq 1$, we have the following:

1. $P/\Phi(P) \cong \mathbb{F}_p^d$ the elementary abelian group of order p^d ;
2. If $N \trianglelefteq P$ and P/N is an elementary abelian group, then $\Phi(P) \leq N$, thus $\Phi(G)$ is the unique normal subgroups with minmal elementary abelian quotient;
3. For $x_1, \dots, x_d \in P$ which equivalence classes $\overline{x_i} \in P/\Phi(P)$ respectively:

$$P = \langle x_1, \dots, x_d \rangle \iff \overline{x_1}, \dots, \overline{x_d} \text{ is a basis for } \mathbb{F}_p^d$$

Proof. (1) We know any maximal subgroup $M < P$ is normal, for a maximal normal subgroup the quotient is simple, so $|P/M| = p$. Thus $P' \leq M$ and $x^p \in M$ for all $x \in P$, so $P' \leq \Phi(P)$ and $P/\Phi(P)$ is elementary abelian.

(2) Suppose $N \trianglelefteq P$ and P/N is elementary abelian, then $\Phi(P/N) = 1$. The maximal subgroups of P/N are of the form M/N for some maximal $M < P$ containing N . As $\Phi(P/N) = N$, the intersection of these must be contained in N , thus $\Phi(P) \leq N$.

(3) If $x_1, \dots, x_d$ span P , then they must span $\mathbb{F}_p^d$ and so form a basis. If $\overline{x_1}, \dots, \overline{x_d}$ is a basis for $\mathbb{F}_p^d$ then $P = \langle x_1, \dots, x_d, \Phi(P) \rangle$ and thus $P = \langle x_1, \dots, x_d \rangle$. $\square$

We can use this to relate the minimal number of generators of P to its Frattini subgroup:

The minimal number of generators of D_8 is 2, so $D_8/\Phi(D_8) \cong C_2^2$, the only order two subgroup $\Phi(D_8)$ can be is $\langle r^2 \rangle$.

3.2 Groups of order p^3

The only ones left are non-abelian groups of order p^3 with p odd.

Lemma 3.2.1. Let G be a group such that $G' \leq Z(G)$, then for all $x, y \in G$ and $n \geq 1$:

$$x^n y^n = (xy)^n [x^{-1}, y^{-1}]^{n(n-1)/2}$$

Proof. We go by induction on n , noting:

$$xy = yx[x^{-1}, y^{-1}]$$

and that by assumption $[x^{-1}, y^{-1}] \in Z(G)$. Assume the claim is true for n and consider:

$$\begin{aligned} (xy)^{n+1} [x^{-1}, y^{-1}]^{n(n+1)/2} &= (xy)^n [x^{-1}, y^{-1}]^{n(n-1)/2} xy [x^{-1}, y^{-1}]^n \\ &= x^n y^n x [x^{-1}, y^{-1}]^n y = x^n y^{n-1} xy [x^{-1}, y^{-1}]^{n-1} y \\ &\vdots \\ &= x^{n+1} y^{n+1} \end{aligned}$$

□

Theorem 3.2.2. For p an odd prime, there are two non-abelian groups of order p^3 up to isomorphism:

1. $P_1 = \langle a, b, c \mid a^p = b^p = c^p = 1, [a, b] = c, [a, c] = [b, c] = 1 \rangle \cong \langle a, c \rangle \rtimes \langle b \rangle \cong C_p^2 \rtimes C_p$;
2. $P_2 = \langle a, b \mid a^{p^2} = b^p = 1, [a, b] = a^p \rangle \cong \langle a \rangle \rtimes \langle b \rangle \cong C_{p^2} \rtimes C_p$.

Proof. Let P be such a group, then By the Burnside basis theorem $P/\Phi(P) \cong C_p^d$ for one of $d = 1, 2, 3$. If $d = 1$, then P is cyclic; if $d = 3$ then $P \cong C_p^3$ abelian, so we must have $d = 2$. Write:

$$P = \langle a, b \rangle, \quad \Phi(P) = \langle c \rangle$$

for some $a, b, c \in P$. We also know $\Phi(P) = Z(P) = P'$.

First, assume $a^p = b^p = 1$. As $[a, b] \neq 1$ (P non-abelian), take $c = [a, b]$, this gives all the relations for P_1 , so this case is done.

Secondly, assume a has order p^2 , from the proof of the Burnside basis theorem we know $a^p \in \Phi(P)$, so $\Phi(P) = \langle a^p \rangle$. If $b^p = 1$, then by taking $[a, b] = a^p$ we have all the relations for P_2 .

If, instead, we let b have order p^2 , wlog we may assume $a^p = c$ and $b^p = c^{-1}$ (by choosing different powers of a and b). Now by the previous lemma we can write

$$1 = a^p b^p = (ab)^p [a^{-1}, b^{-1}]^{p(p-1)/2}$$

where p being odd must divide $p(p-1)/2$, thus $[a^{-1}, b^{-1}]^{p(p-1)/2} = 1$ and hence $(ab)^p = 1$ and by replacing b with ab we reach the first state. □

P_1 also has an upper unitriangular representation over $\mathbb{F}_p^3$ given by:

$$a \mapsto \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad b \mapsto \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \quad c \mapsto \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

3.3 Groups of order 16

Not lectured.

Definition 3.3.1. Here are four new classes of groups of order p^n for $n \geq 3$:

- Mod_{p^n}
- D_{2^n}
- SD_{2^n}
- Q_{2^n}

3.4 The transfer homomorphism